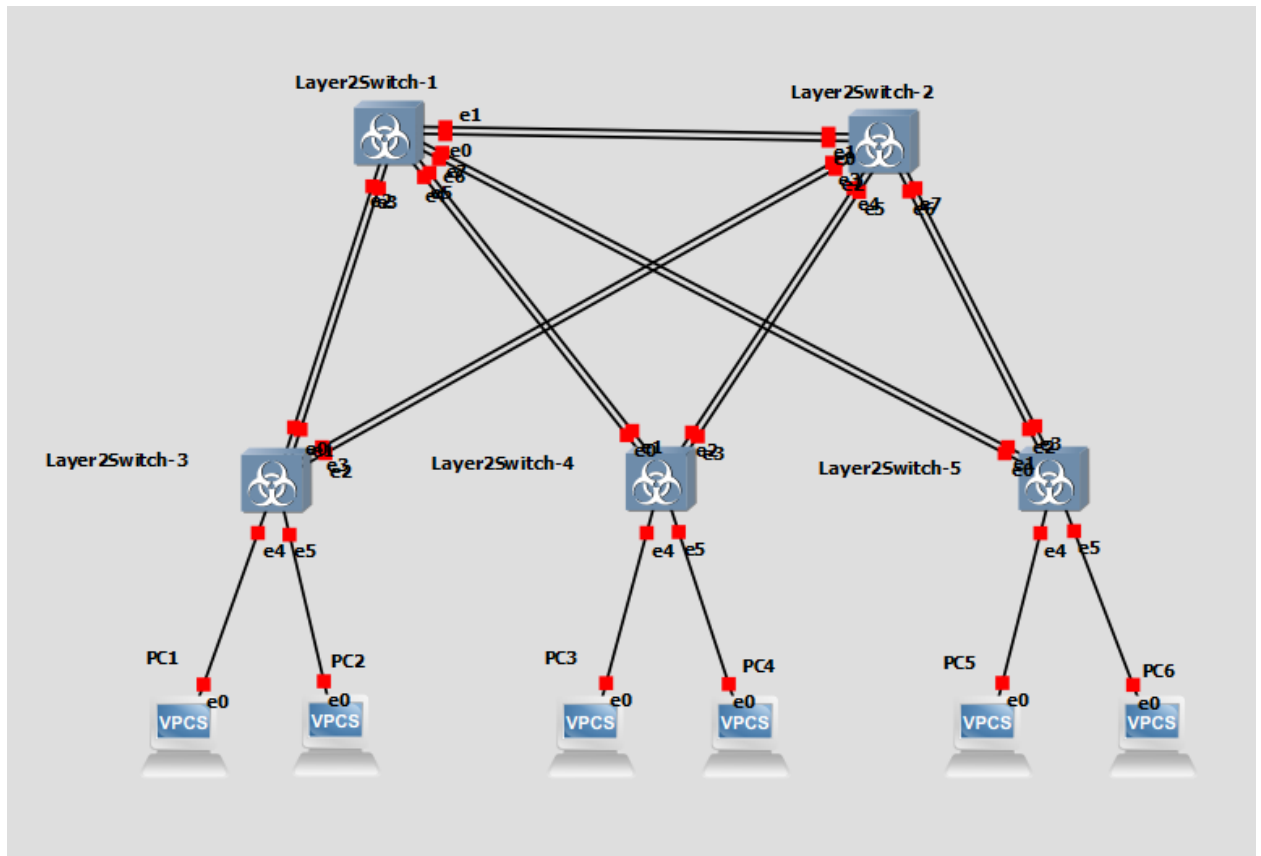


Тема: Настройка протокола STP (IEEE 802.1D)



- 1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета.

На **первом** свитче:

```
enable
conf t
spa vlan 1 prio 28672
end
write
```

Остальные свитчи оставляем по умолчанию(32768).

```

vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spa vlan 1 prio 30000
% Bridge Priority must be in increments of 4096.
% Allowed values are:
   0      4096   8192   12288   16384   20480   24576   28672
 32768  36864  40960  45056  49152  53248  57344  61440
vIOS-L2-01(config)#spa vlan 1 prio 28672
vIOS-L2-01(config)#end
vIOS-L2-01#w
*May 25 18:06:34.096: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write

```

На втором свитче изменился root id:

```

Layer2Switch-2 - PuTTY
vIOS-L2-01>show spa

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    28673
            Address    0cd9.204a.0000
            Cost        4
            Port        1 (GigabitEthernet0/0)
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
            Address    0c60.f5f1.0000
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time   300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Root FWD 4        128.1   Shr
Gi0/1          Altn BLK 4        128.2   Shr
Gi0/2          Desg FWD 4        128.3   Shr
Gi0/3          Desg FWD 4        128.4   Shr
Gi1/0          Desg FWD 4        128.5   Shr
Gi1/1          Desg FWD 4        128.6   Shr
Gi1/2          Altn BLK 4        128.7   Shr
Gi1/3          Altn BLK 4        128.8   Shr
Gi2/0          Desg FWD 4        128.9   Shr

vIOS-L2-01>

```

- 2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать.

Прописываем ip адреса для каждого VPCS:

192.168.1.n – где n номер vpcs.

```
PC6 startup.vpc

# This the configuration for PC6
#
# Uncomment the following line to enable DHCP
# dhcp
# or the line below to manually setup an IP address and subnet mask
ip 192.168.1.6 255.255.255.0
#

set pcname PC6
```

Пинги:

```
PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=3.211 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.227 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.890 ms
^C
PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=8.260 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=9.033 ms
^C
PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=12.917 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=4.107 ms
^C
PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=5.291 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=9.917 ms
^C
PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=8.065 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=4.597 ms
^C
PC1> █
```

```
PC2> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=5.405 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.575 ms
^C
PC2> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=11.760 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=11.313 ms
^C
PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=9.300 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=9.748 ms
^C
PC2>
PC2> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=13.904 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=7.223 ms
^C
PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=5.456 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=12.128 ms
^C
PC2> █
```

```
PC3> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=17.649 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=8.422 ms
^C
PC3> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=20.134 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=8.325 ms
^C
PC3> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=3.143 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=2.533 ms
^C
PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=20.516 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=9.061 ms
^C
PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=10.206 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=15.820 ms
^C
PC3> █
```

```
PC4> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=6.324 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=5.642 ms
^C
PC4> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=7.803 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=10.351 ms
^C
PC4> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.603 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=11.500 ms
^C
PC4> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=10.735 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=10.512 ms
^C
PC4> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=8.022 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=10.473 ms
^C
PC4> █
```

```

PC5>
PC5> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=15.091 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=5.280 ms
^C
PC5> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=6.609 ms
^C
PC5> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=14.510 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=7.345 ms
^C
PC5> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=9.670 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=12.197 ms
^C
PC5> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=7.153 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=0.687 ms
^C
PC5>

```

```

PC6> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=15.002 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=5.636 ms
^C
PC6> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=3.547 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=8.096 ms
^C
PC6> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=8.155 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=8.367 ms
^C
PC6> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=7.708 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=6.080 ms
^C
PC6> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=0.837 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=3.622 ms
^C
PC6>

```

- 3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл.

Через show spa находим информацию:

```

Layer2Switch-5 - PuTTY
vIOS-L2-01>show spa

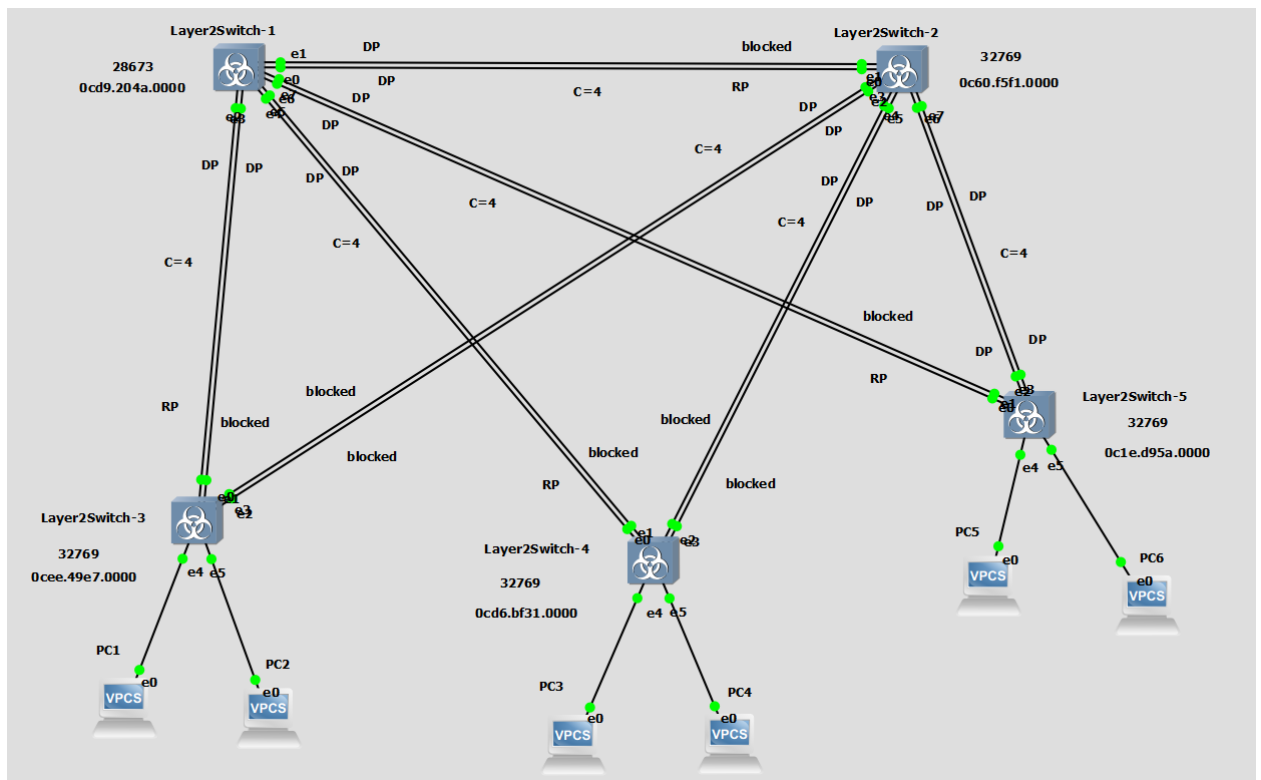
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    28673
             Address     0cd9.204a.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0cle.d95a.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface          Role Sts Cost      Prio.Nbr Type
-----
Gi0/0              Root FWD 4         128.1   Shr
Gi0/1              Altn BLK 4         128.2   Shr
Gi1/0              Desg FWD 4         128.5   Shr
Gi1/1              Desg FWD 4         128.6   Shr

```

Аналогично делаем с другими свитчами.



(в отдельном файле – пункт3.png)

- 4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет.

Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 28672/1/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
2	0.949851	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/100/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
3	0.953788	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/200/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
4	0.979120	0c:60:f5:f1:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/100/0c:60:f5:f1:00:00 Cost = 0 Port = 0x8002
5	0.989743	0c:60:f5:f1:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/200/0c:60:f5:f1:00:00 Cost = 0 Port = 0x8002
6	0.993865	0c:60:f5:f1:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/300/0c:60:f5:f1:00:00 Cost = 0 Port = 0x8002
7	1.009746	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 28672/1/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
8	1.028818	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/300/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
9	2.019559	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 28672/1/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
10	2.973542	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/100/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
11	2.978466	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/200/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
12	3.022465	0c:60:f5:f1:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/100/0c:60:f5:f1:00:00 Cost = 0 Port = 0x8002
13	3.032520	0c:60:f5:f1:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/200/0c:60:f5:f1:00:00 Cost = 0 Port = 0x8002
14	3.034771	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 28672/1/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
15	3.037145	0c:60:f5:f1:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/300/0c:60:f5:f1:00:00 Cost = 0 Port = 0x8002
16	3.053749	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/200/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
17	4.059319	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 28672/1/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002
18	5.022310	0c:d9:20:4a:00:01	Nearest-Customer-Bridge	STP	60	Conf., Root = 32768/100/0c:d9:20:4a:00:00 Cost = 0 Port = 0x8002

Отправка от Layer2Switch-(root)1 до Layer2Switch2:

```
IEEE 802.3 Ethernet
  Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
  Source: 0c:d9:20:4a:00:01 (0c:d9:20:4a:00:01)
  Length: 38
  Padding: 0000000000000000
  [Stream index: 0]
  Logical-Link Control
    DSAP: Spanning Tree BPDU (0x42)
    SSAP: Spanning Tree BPDU (0x42)
    Control field: U, func=UI (0x03)
  Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
    BPDU flags: 0x00
    Root Identifier: 28672 / 1 / 0c:d9:20:4a:00:00
    Root Path Cost: 0
    Bridge Identifier: 28672 / 1 / 0c:d9:20:4a:00:00
    Port identifier: 0x8002
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

Ретрансляция hello от root(от Layer2Switch-2 до Layer2Switch-5):

```
Frame 11: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
IEEE 802.3 Ethernet
  Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
  Source: 0c:60:f5:f1:00:06 (0c:60:f5:f1:00:06)
  Length: 38
  Padding: 0000000000000000
  [Stream index: 0]
  Logical-Link Control
  Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
    BPDU flags: 0x00
    Root Identifier: 28672 / 1 / 0c:d9:20:4a:00:00
    Root Path Cost: 4
    Bridge Identifier: 32768 / 1 / 0c:60:f5:f1:00:00
    Port identifier: 0x8007
    Message Age: 1
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

Аналогичная передача Hello-пакетов наблюдается на всех остальных линках сети. Корневой коммутатор генерирует Hello-пакеты каждые 2 секунды (Hello Time = 2) с Message Age = 0 на всех своих активных портах. Некорневые коммутаторы (Layer2Switch-2, 3, 4, 5) ретранслируют полученные BPDU на своих Designated-портах с увеличенным Message Age, обеспечивая доставку информации о топологии до всех устройств сети. Заблокированные порты Hello-пакеты получают, но не передают.

- 5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл.

Было:

```
Layer2Switch-3 - PuTTY
vIOS-L2-01#show spa

VLAN0001
  Spanning tree enabled protocol ieee
    Root ID    Priority    28673
              Address    0cd9.204a.0000
              Cost        4
              Port        1 (GigabitEthernet0/0)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
              Address    0cee.49e7.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   300 sec

Interface      Role  Sts  Cost      Prio.Nbr  Type
-----
Gi0/0          Root  FWD  4          128.1     Shr
Gi0/1          Altn  BLK  4          128.2     Shr
Gi0/2          Altn  BLK  4          128.3     Shr
Gi0/3          Altn  BLK  4          128.4     Shr
Gi1/0          Desg  FWD  4          128.5     Shr
Gi1/1          Desg  FWD  4          128.6     Shr
```

Enable
conf t
int Gi0/0
spa cost 15
end
write

Стало:

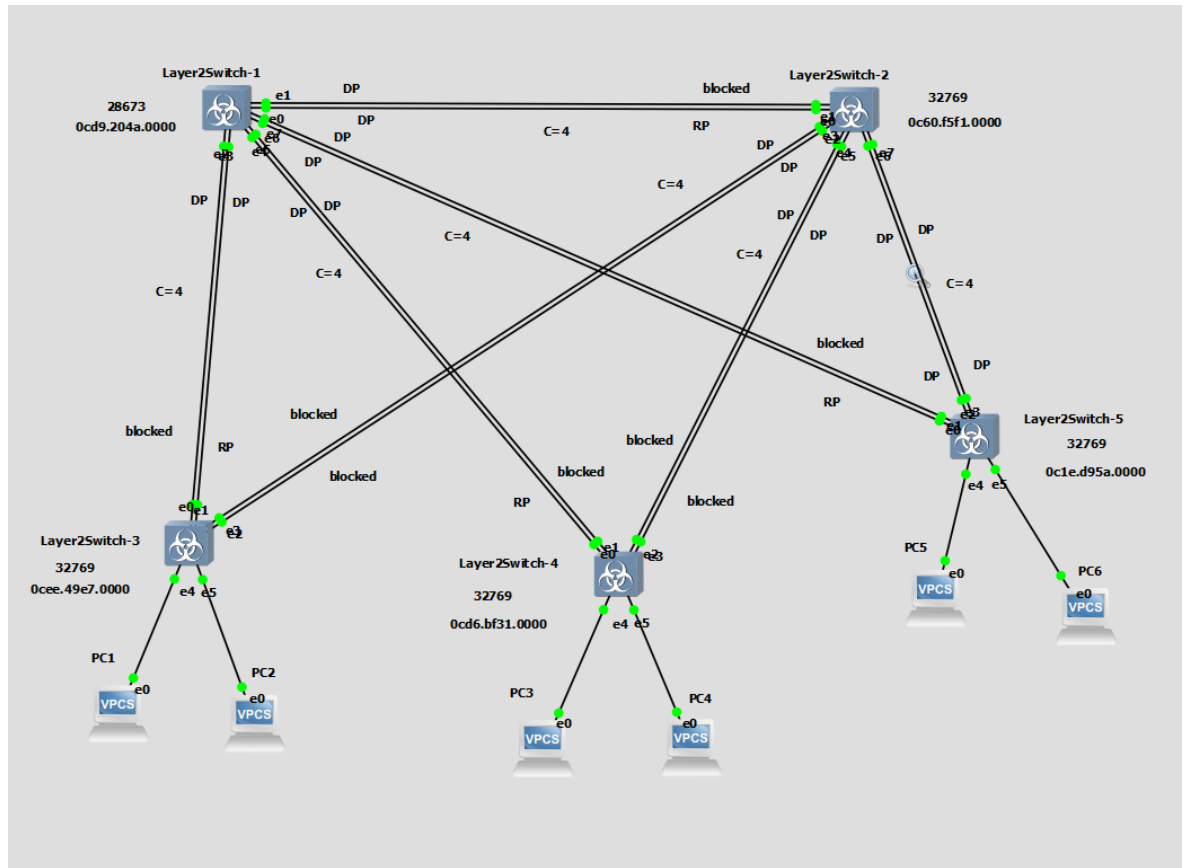
```
disk successfully.
vIOS-L2-01#show spa

VLAN0001
  Spanning tree enabled protocol ieee
    Root ID    Priority    28673
              Address    0cd9.204a.0000
              Cost        4
              Port        2 (GigabitEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
              Address    0cee.49e7.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   300 sec

Interface      Role  Sts  Cost      Prio.Nbr  Type
-----
Gi0/0          Altn  BLK  15         128.1     Shr
Gi0/1          Root  FWD  4          128.2     Shr
Gi0/2          Altn  BLK  4          128.3     Shr
Gi0/3          Altn  BLK  4          128.4     Shr
Gi1/0          Desg  FWD  4          128.5     Shr
Gi1/1          Desg  FWD  4          128.6     Shr
```

Результат:



RP и blocked port поменялись местами.